

LECTURE 6: MATH1061/7861
 TOPICS: QUANTIFIED STATEMENTS AND THEIR
 NEGATIONS
 DATE: THURSDAY MARCH 5, 2026

Learning objectives:

- (1) Gain fluency in translating quantified statements to and from English statements.
- (2) Be able to determine if a quantified statement is true or false.

Notation:

• $\mathbb{N}$ Natural numbers $\{0, 1, 2, \dots\}$

• $\mathbb{Z}$ integers $\{\dots, -3, -1, 0, 1, 2, \dots\}$

• $\mathbb{Q}$ Rational numbers $\left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0 \right\}$

• $\mathbb{R}$ Real numbers measuring distances. Have decimal expansion

• $\mathbb{R} \setminus \mathbb{Q} = \mathbb{R} - \mathbb{Q} = \overline{\mathbb{Q}}$ $\rightarrow$ irrational.

• $\forall$ $\equiv$ For all / For every

• $\exists$ $\equiv$ There exists.

Q1:

Translate the following to quantified statements and determine if they are true or false.

- (1) There is a real number that is an integer.

True.

$$\exists r \in \mathbb{R} \quad \underbrace{\text{s.t.}}_{\text{such that}} \quad r \in \mathbb{Z}$$

- (2) If a real number is an integer, then it is rational.

$$\forall x \in \mathbb{R}, \quad x \in \mathbb{Z} \longrightarrow x \in \mathbb{Q}$$

True. Indeed if $x \in \mathbb{Z}$, then

$$x = \frac{x}{1} \in \mathbb{Q}$$

Q2:

Write the negations of the following statements. Determine if the original statement is true or false. ~~Write the original statement in English.~~

~~(1) $\forall x \in \mathbb{Z}, x \in \mathbb{Q}$~~

s.t.

~~(1)~~ $\exists x \in \mathbb{Q}, \frac{x}{2} \in \mathbb{Q}$

True ✓

$$\frac{1}{2} \in \mathbb{Q}, \quad \frac{\left(\frac{1}{2}\right)}{2} = \frac{1}{4} \in \mathbb{Q}$$

Negation: $\forall x \in \mathbb{Q}, \frac{x}{2} \notin \mathbb{Q}$.

$$\equiv \forall x \in \mathbb{Q}, \frac{x}{2} \in \underbrace{\mathbb{R} - \mathbb{Q}}_{\text{irrational numbers}} = \overline{\mathbb{Q}}$$

$$\mathbb{Z}_{\text{odd}} = \{\dots, -3, -1, 1, 3, 5, \dots\}$$

$$(3) \forall x \in \mathbb{Z}, x^2 \in \mathbb{Z}_{\text{odd}} \rightarrow x \in \mathbb{Z}_{\text{odd}} \quad \text{True}$$

Why?

Negation: $\exists x \in \mathbb{Z}, x^2 \in \mathbb{Z}_{\text{odd}} \wedge x \notin \mathbb{Z}_{\text{odd}}$

$$\equiv \exists x \in \mathbb{Z}, x^2 \in \mathbb{Z}_{\text{odd}} \wedge x \in \mathbb{Z}_{\text{even}}$$

$$(4) \forall x, y \in \mathbb{R} - \mathbb{Q}, xy \in \mathbb{R} - \mathbb{Q}$$

False. $\sqrt{2} \in \mathbb{R} - \mathbb{Q}$

$$\sqrt{2} \cdot \sqrt{2} = 2 \in \mathbb{Q}.$$

Negation: $\exists x, y \in \mathbb{R} - \mathbb{Q}, xy \notin \mathbb{R} - \mathbb{Q}$

$$\equiv \exists x, y \in \mathbb{R} - \mathbb{Q}, xy \in \mathbb{Q}$$

Want to prove:

$$\forall x \in \mathbb{Z}, x^2 \in \mathbb{Z}_{\text{odd}} \longrightarrow x \in \mathbb{Z}_{\text{odd}}$$

Easier to prove the contrapositive:

$$\forall x \in \mathbb{Z}, x \notin \mathbb{Z}_{\text{odd}} \longrightarrow x^2 \notin \mathbb{Z}_{\text{odd}}$$

$$\equiv \forall x \in \mathbb{Z}, x \in \mathbb{Z}_{\text{even}} \longrightarrow x^2 \in \mathbb{Z}_{\text{even}}$$

Indeed $x \in \mathbb{Z}_{\text{even}}$ means $x = 2k$ for some $k \in \mathbb{Z}$.

Then $x^2 = (2k)(2k) = 4k^2 = 2(2k^2)$.
So $x^2 = 2 \times (\text{some integer})$. Thus, $x^2 \in \mathbb{Z}_{\text{even}}$.